U.S. Department of the Interior / U.S. Geological Survey



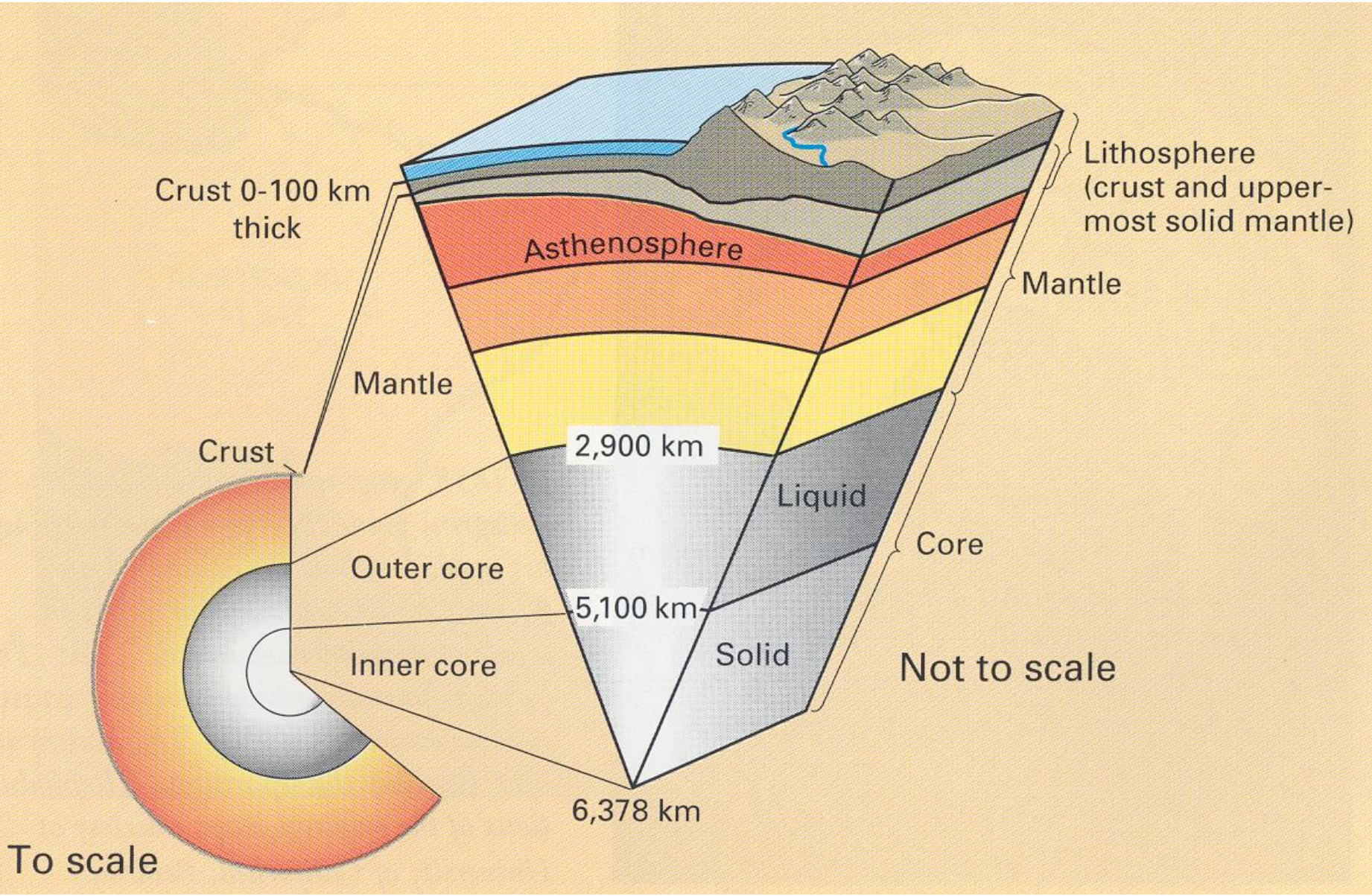
This Dynamic Earth:

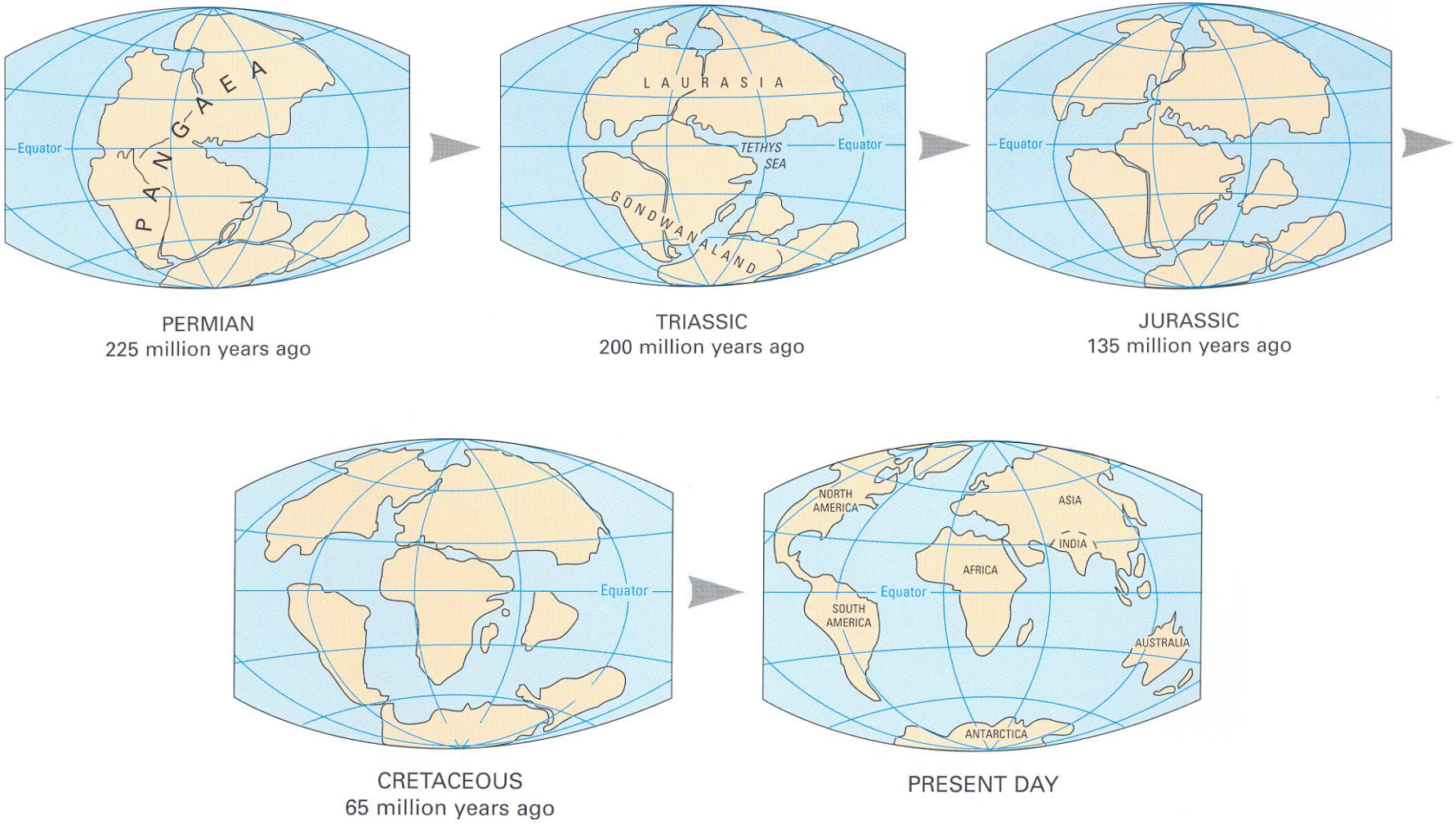
the Story of Plate Tectonics

by
W. Jacquelyne Kious and
Robert I. Tilling

Book designed and coordinated by Martha Kiger
Illustrations and production by Jane Russell

This Dynamic Earth:
the Story of Plate Tectonics


Crust 0-100 km thick
Lithosphere (crust and upper-most solid mantle)
Asthenosphere
Mantle
Mantle
Crust
2,900 km
Liquid
Core
Outer core
5,100 km
Solid
Inner core
Not to scale
6,378 km
To scale



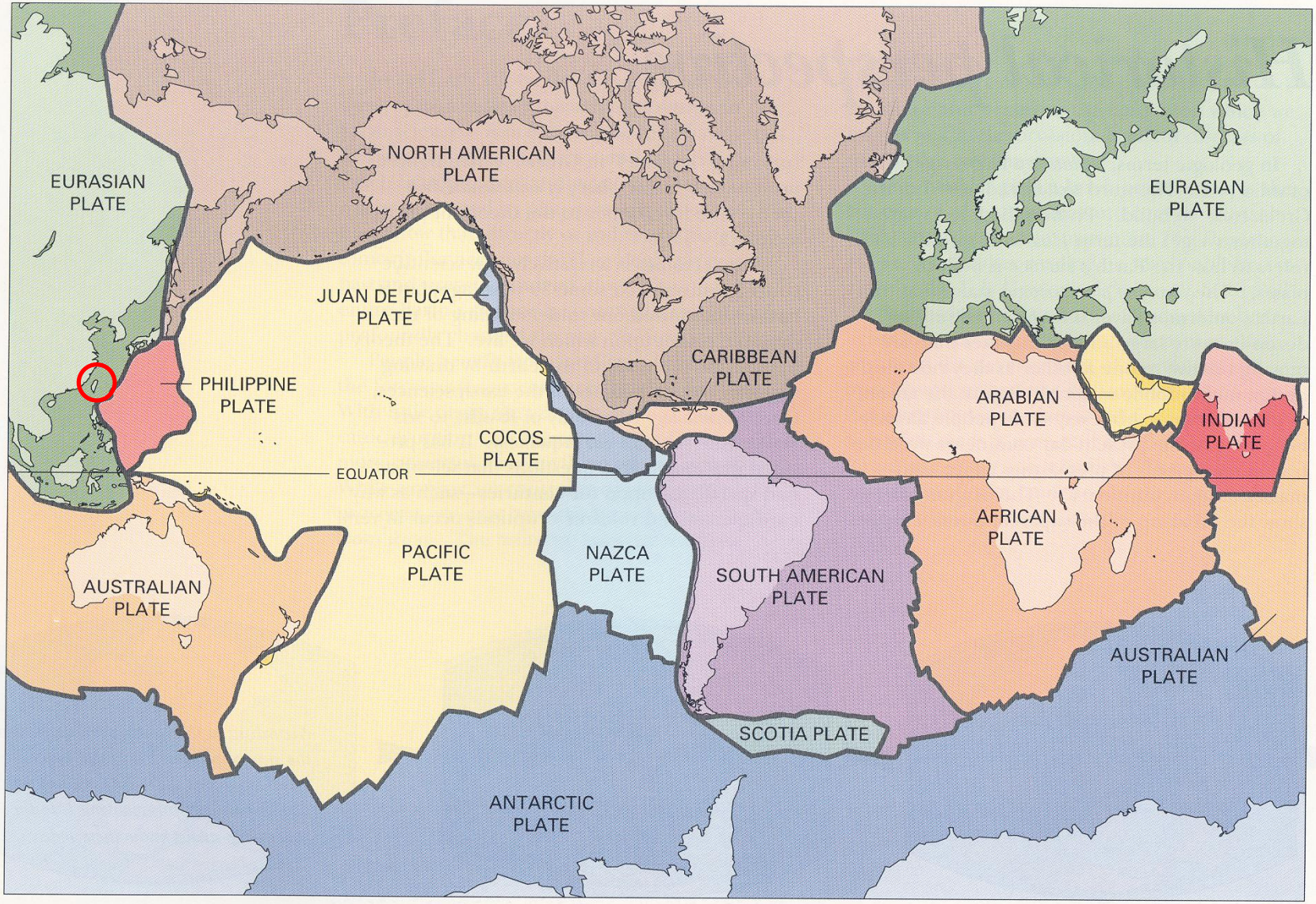
PERMIAN
225 million years ago

TRIASSIC
200 million years ago

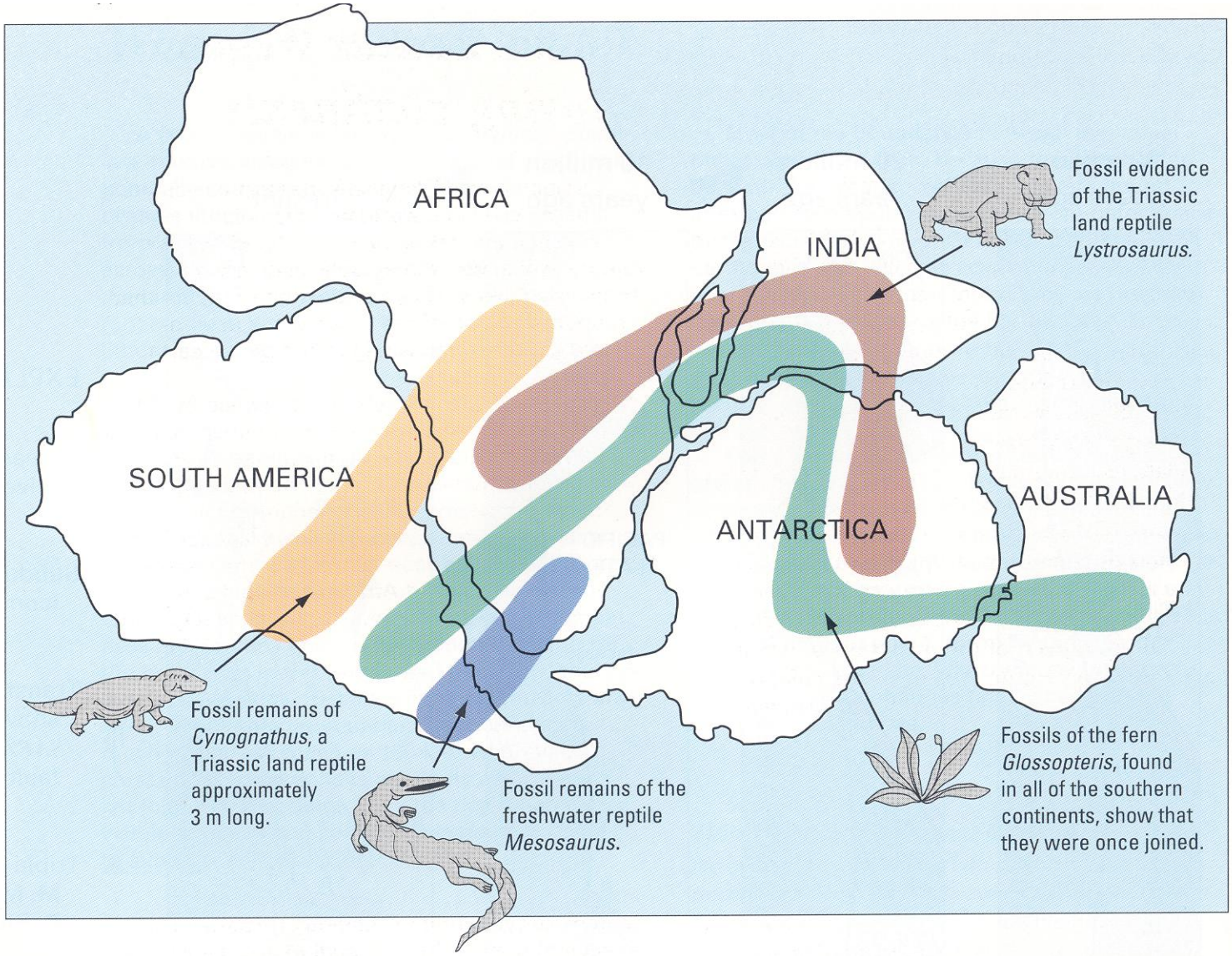
JURASSIC
135 million years ago

CRETACEOUS
65 million years ago

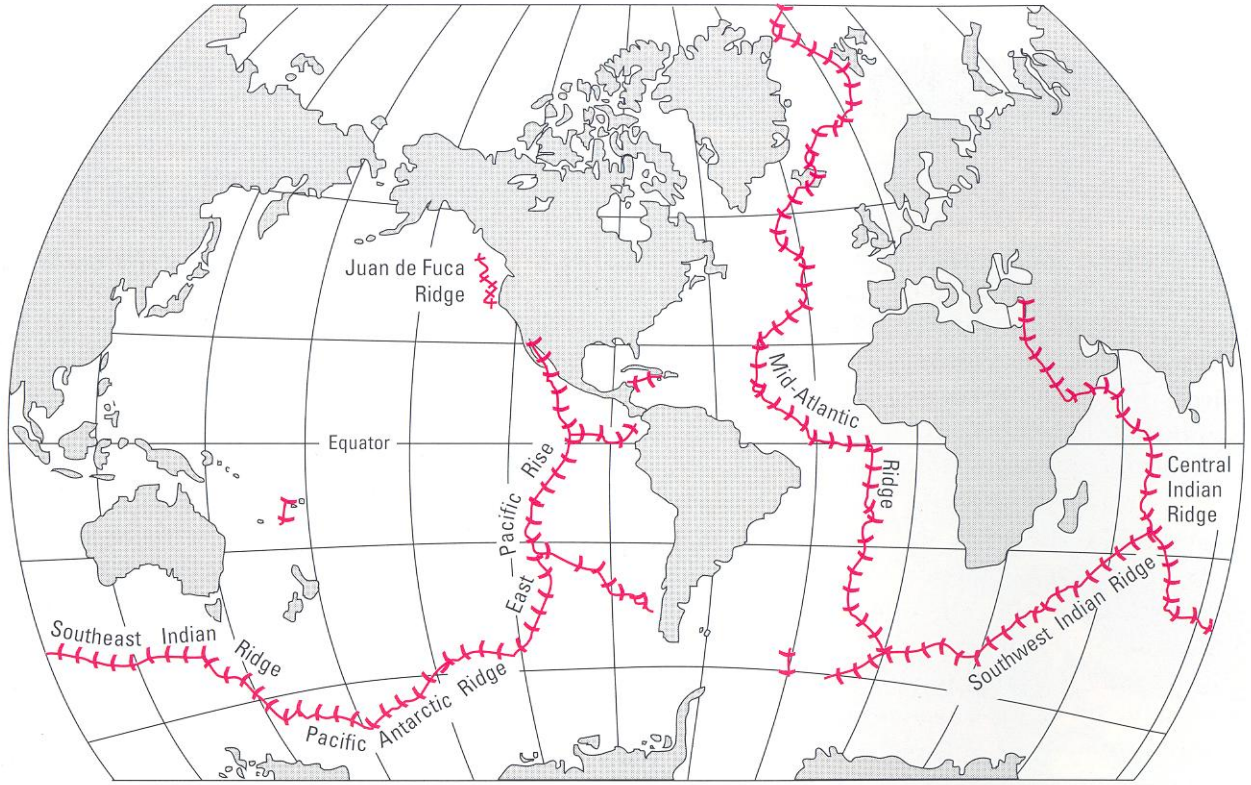
PRESENT DAY

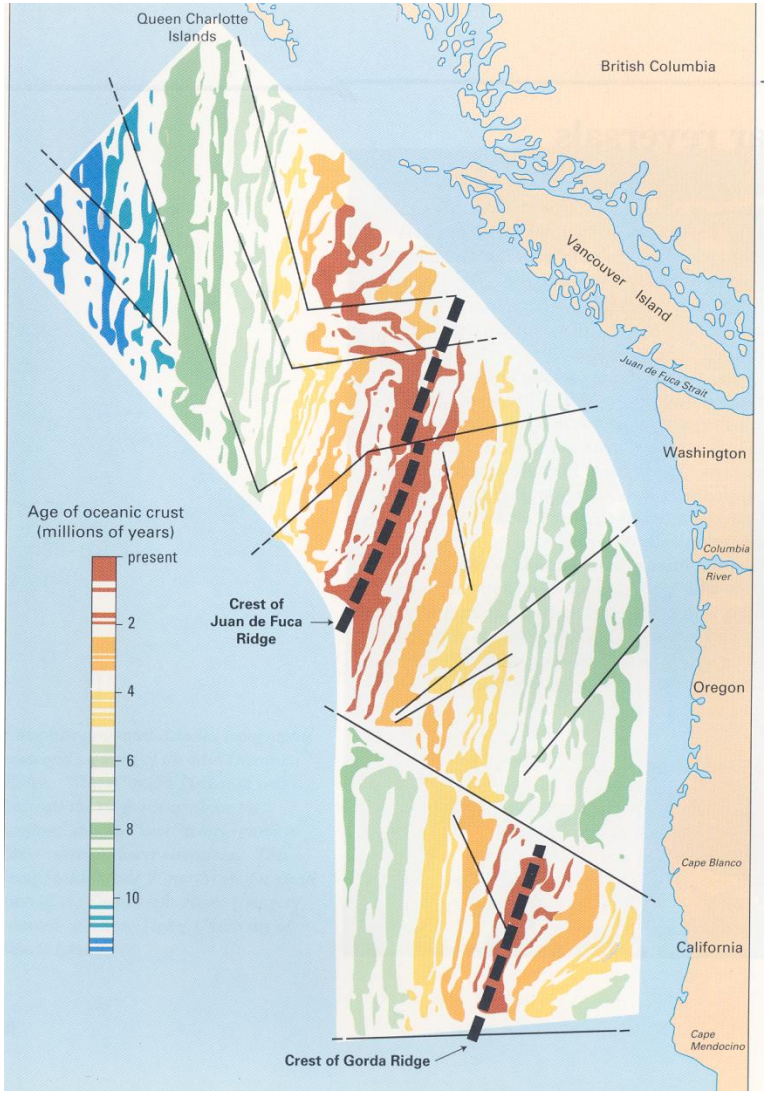


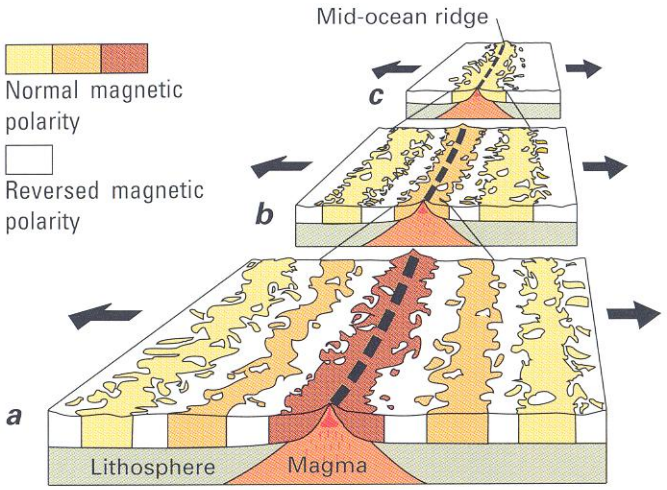
The layer of the Earth we live on is broken into a dozen or so rigid slabs (called tectonic plates *by geologists) that are moving relative to one another.*

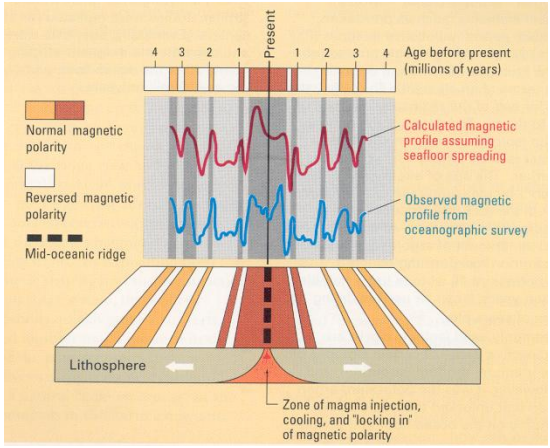


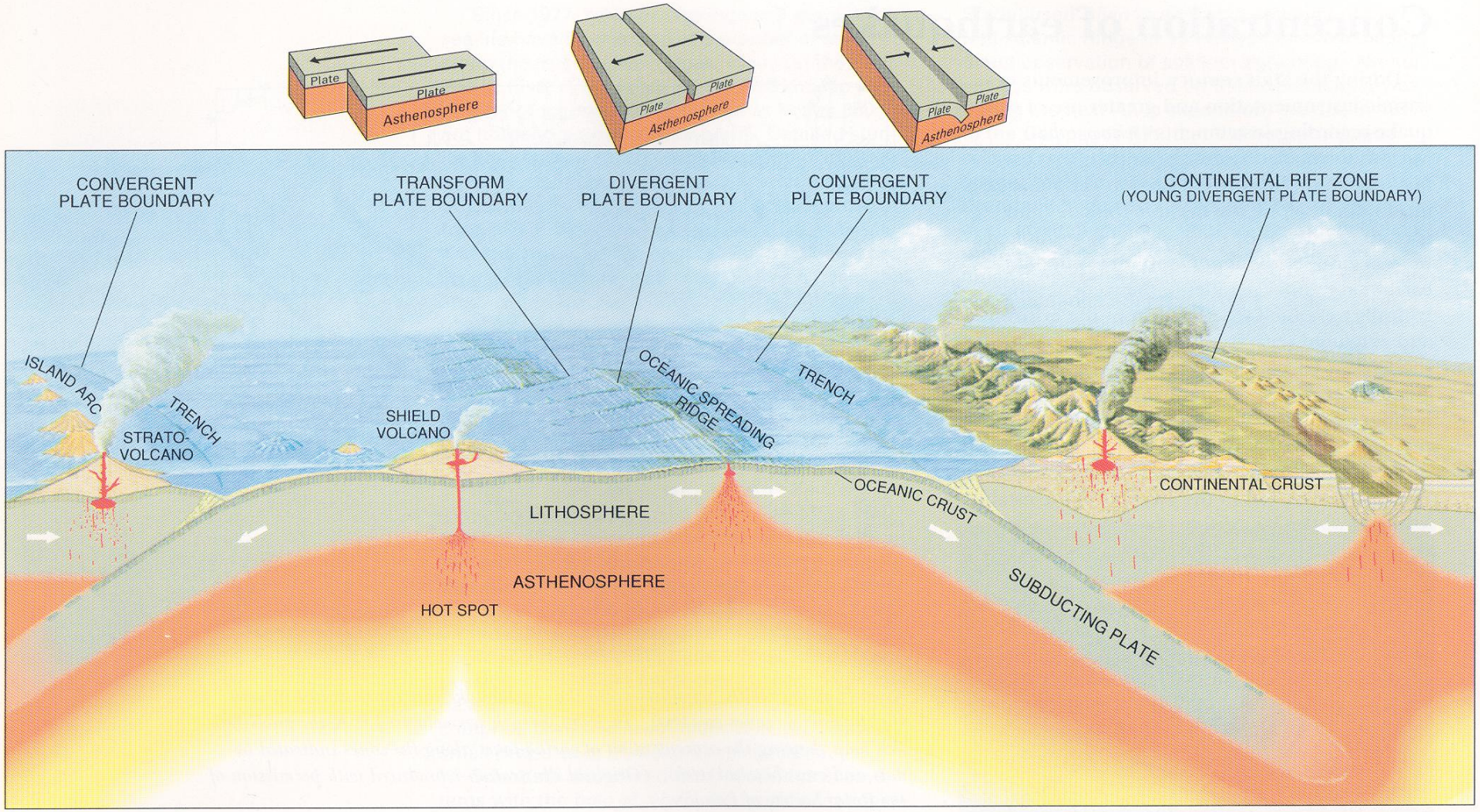
As noted by Snider-Pellegrini and Wegener, the locations of certain fossil plants and animals on present-day, widely separated continents would form definite patterns (shown by the bands of colors), if the continents are rejoined.


Juan de Fuca Ridge
Equator
East Pacific Rise
Mid-Atlantic Ridge
Central Indian Ridge
Southeast Indian Ridge
Pacific Antarctic Ridge
Southwest Indian Ridge

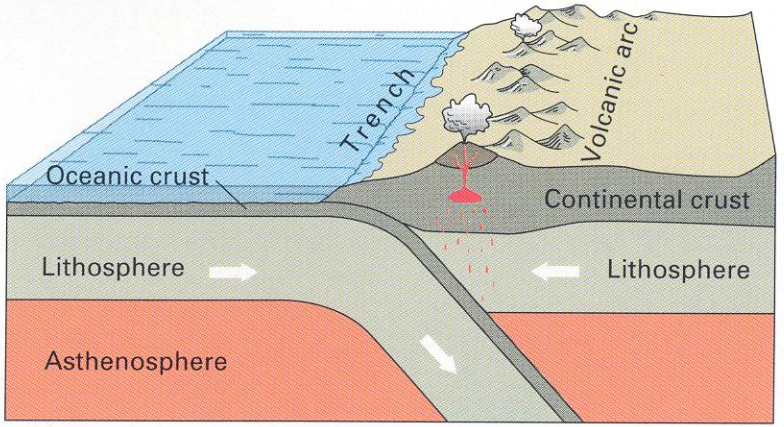

Queen Charlotte Islands
British Columbia
Vancouver Island
Juan de Fuca Strait
Washington
Columbia River
Oregon
Cape Blanco
California
Cape Mendocino
Age of oceanic crust (millions of years)
present
2
4
6
8
10
Crest of Juan de Fuca Ridge
Crest of Gorda Ridge


Mid-ocean ridge
Normal magnetic polarity
Reversed magnetic polarity
a
b
c
Lithosphere
Magma

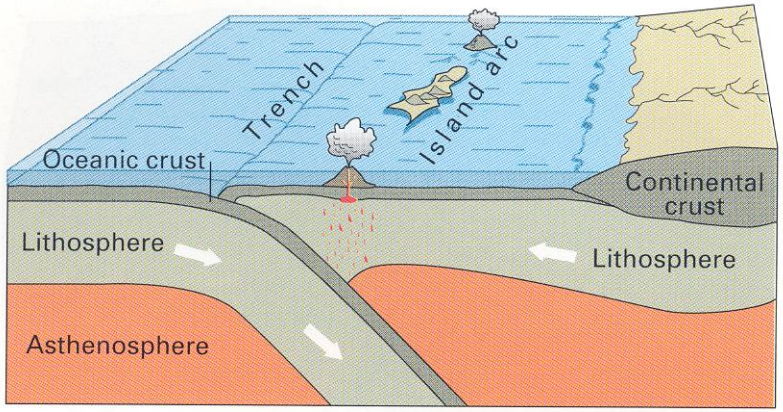

Present
4 3 2 1
1 2 3 4
Age before present (millions of years)
Normal magnetic polarity
Reversed magnetic polarity
Mid-oceanic ridge
Calculated magnetic profile assuming seafloor spreading
Observed magnetic profile from oceanographic survey
Lithosphere
Zone of magma injection, cooling, and "locking in" of magnetic polarity



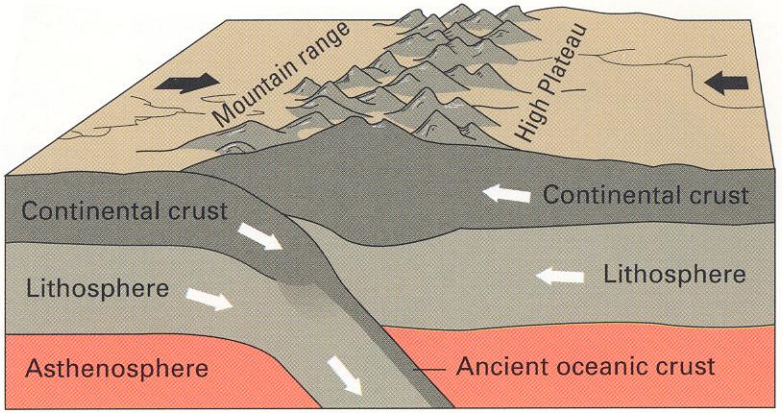
Artist's cross section illustrating the main types of plate boundaries (see text); East African Rift Zone is a good example of a continental rift zone. (Cross section by José F. Vigil from This Dynamic Planet*—a wall map produced jointly by the U.S. Geological Survey, the Smithsonian Institution, and the U.S. Naval Research Laboratory.)*



Oceanic-continental convergence



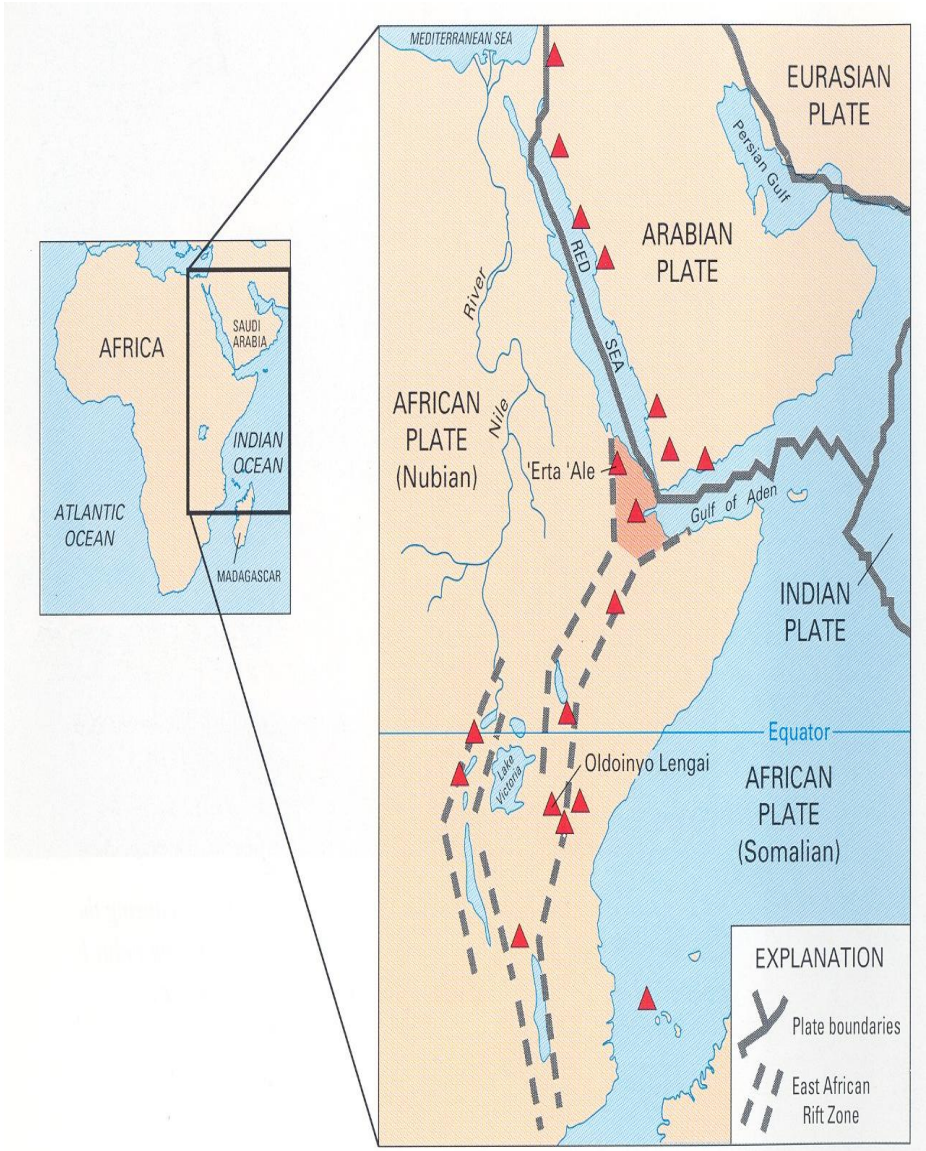
Oceanic-oceanic convergence



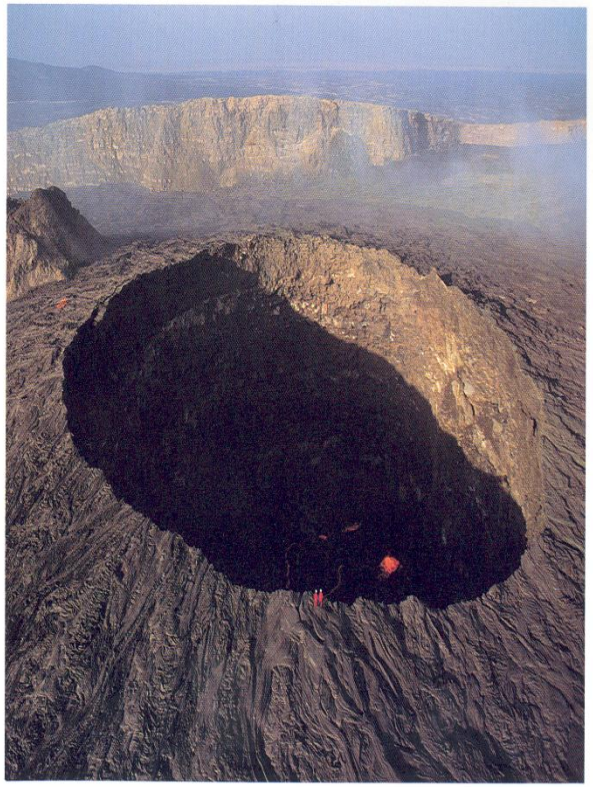
Continental-continental convergence

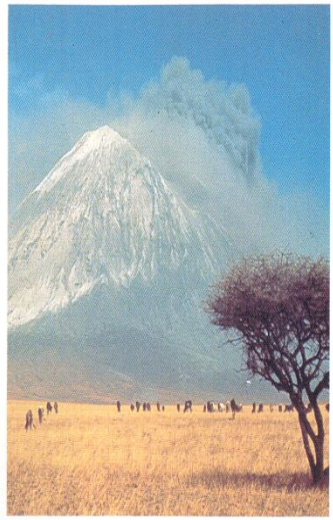

ICELAND
RIDGE
AFRICAN
Equator
AMERICAN
LATE
MID-ATLANTIC



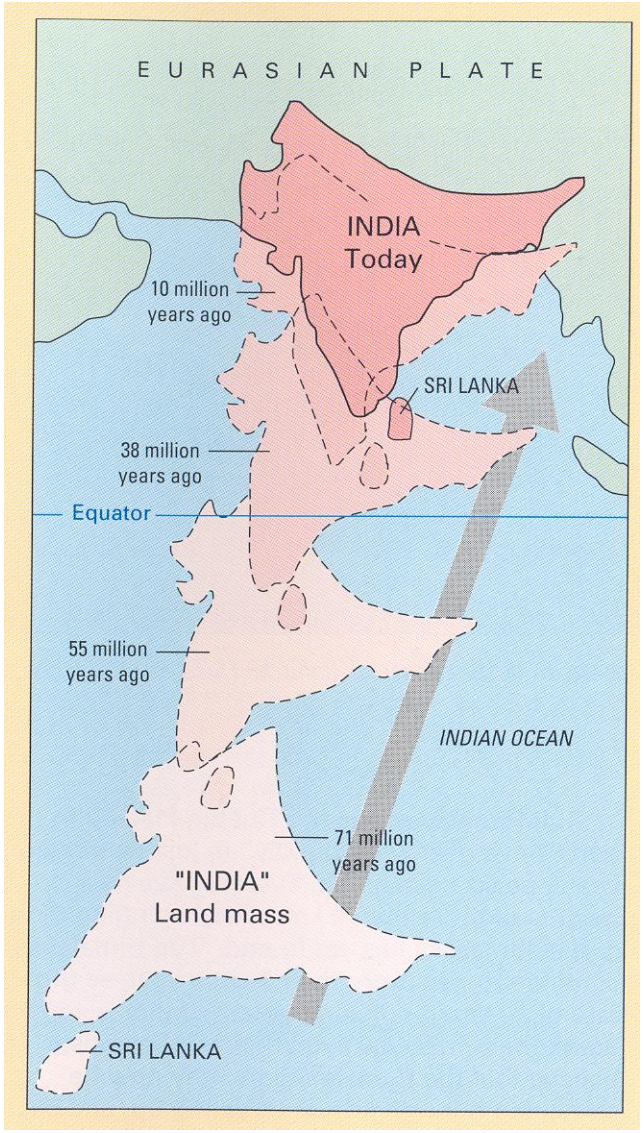


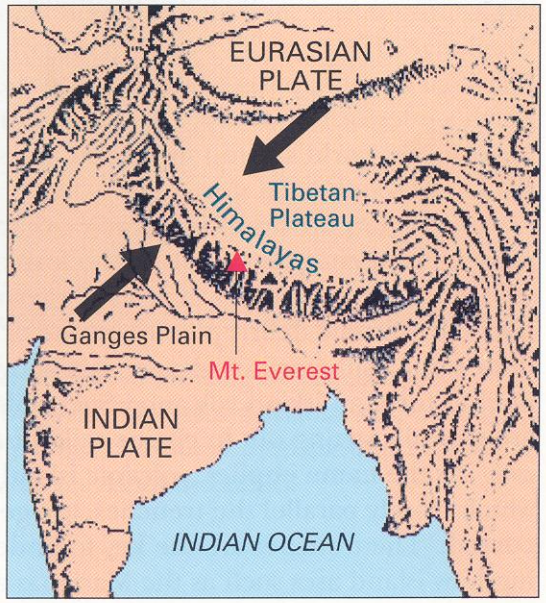
Helicopter view (in February 1994) of the active lava lake within the summit crater of 'Erta 'Ale (Ethiopia), one of the active volcanoes in the East African Rift Zone. Two helmeted, red-suited volcanologists —observing the activity from the crater rim—provide scale. Red color within the crater shows where molten lava is breaking through the lava lake's solidified, black crust. (Photograph by Jacques Durieux, Groupe Volcans Actifs.)

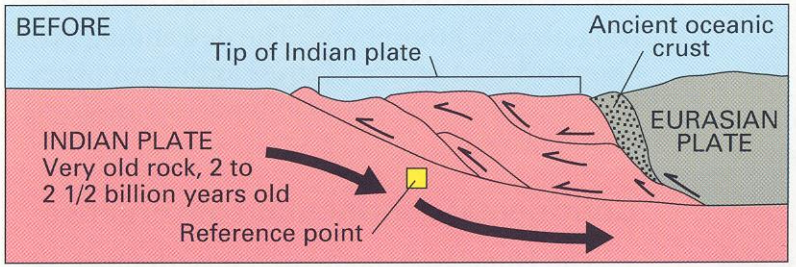


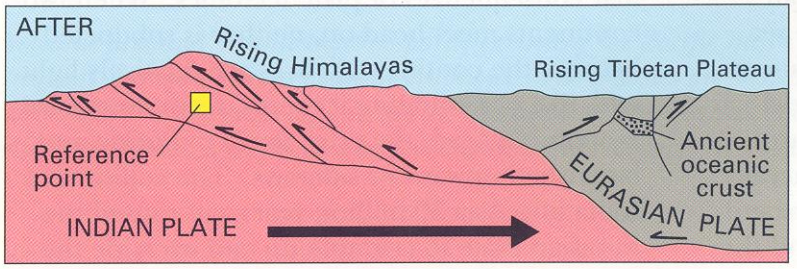


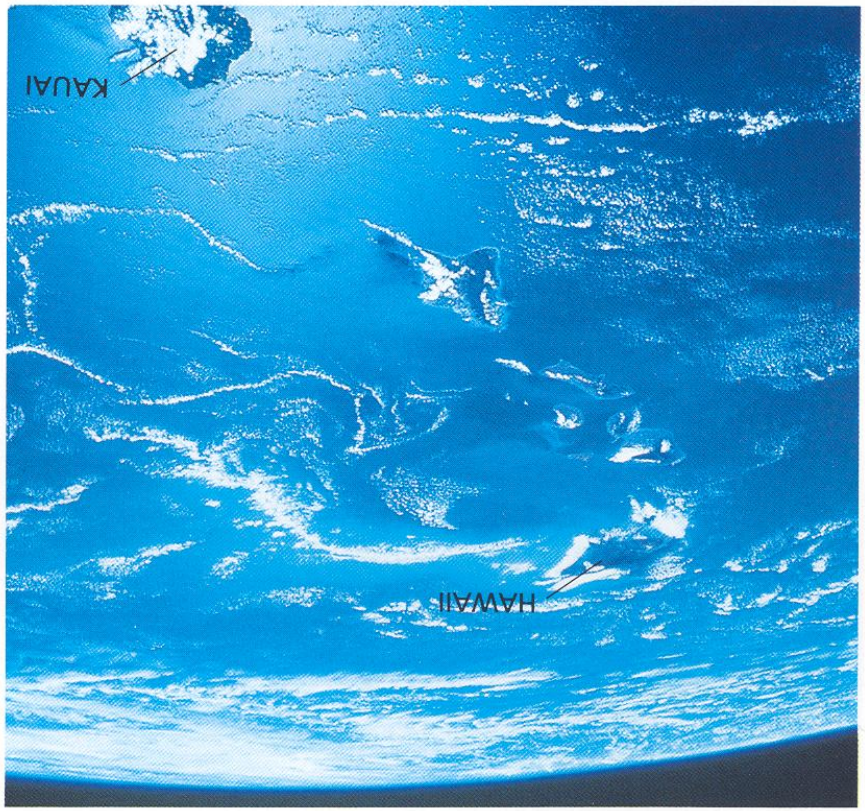
Oldoinyo Lengai, another active volcano in the East African Rift Zone, erupts explosively in 1966. (Photograph by Gordon Davies, courtesy of Celia Nyamweru, St. Lawrence University, Canton, New York.)

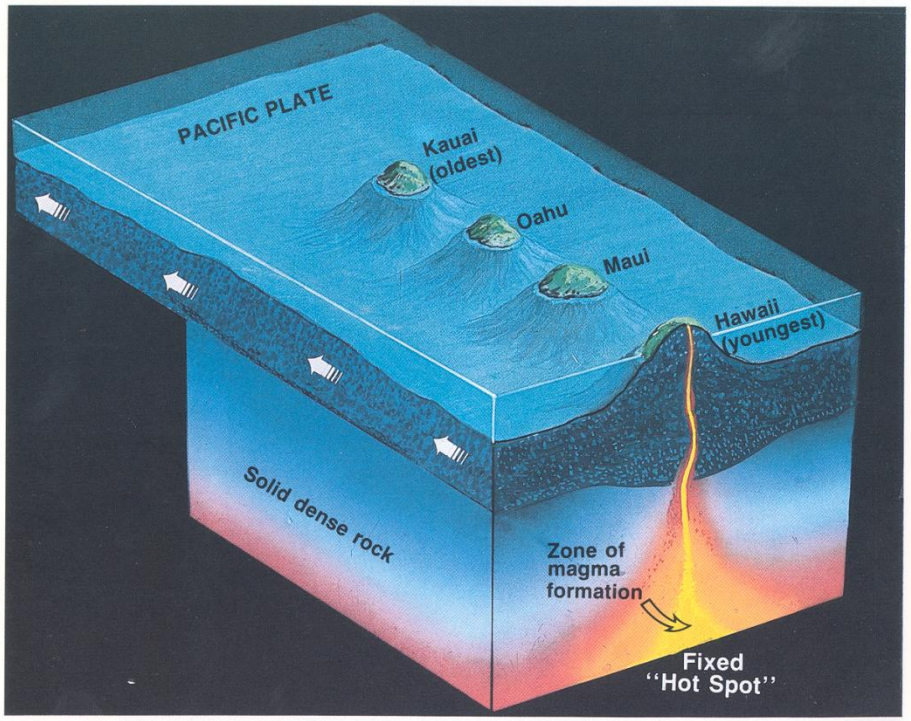

EURASIAN PLATE
INDIA
Today
10 million
years ago
SRI LANKA
38 million
years ago
Equator
55 million
years ago
INDIAN OCEAN
71 million
years ago
"INDIA"
Land mass
SRI LANKA

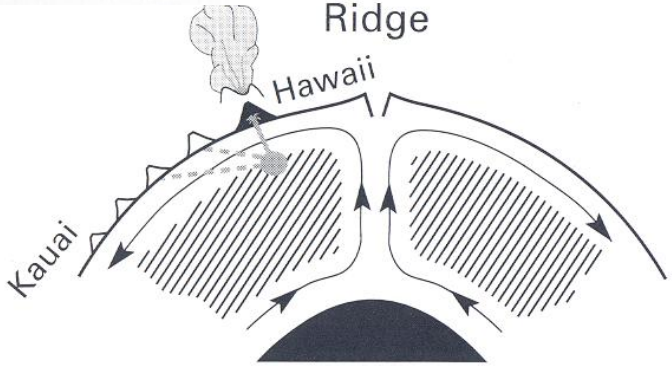

EURASIAN
PLATE
Tibetan
Plateau
Himalayas
Ganges Plain
Mt. Everest
INDIAN
PLATE
INDIAN OCEAN


BEFORE
Tip of Indian plate
Ancient oceanic
crust
EURASIAN
PLATE
INDIAN PLATE
Very old rock, 2 to
2 1/2 billion years old
Reference point


AFTER
Rising Himalayas
Rising Tibetan Plateau
Reference
point
Ancient
oceanic
crust
EURASIAN PLATE
INDIAN PLATE



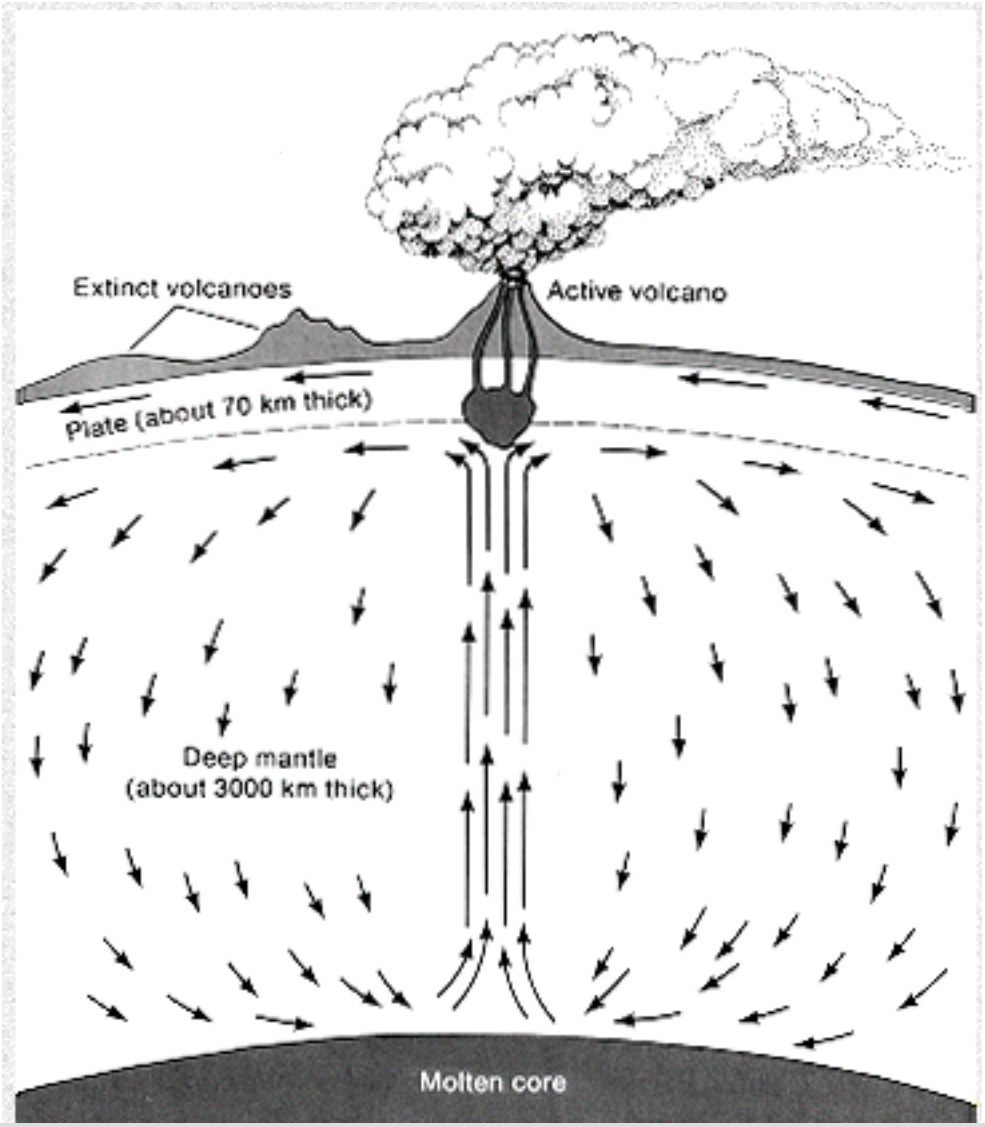




(*a*) Section



(*b*) Plan



Deep mantle plumes generated by slow convection currents are one explanation of the relatively fixed position of hot-spot volcanic sources.

EXPLANATION

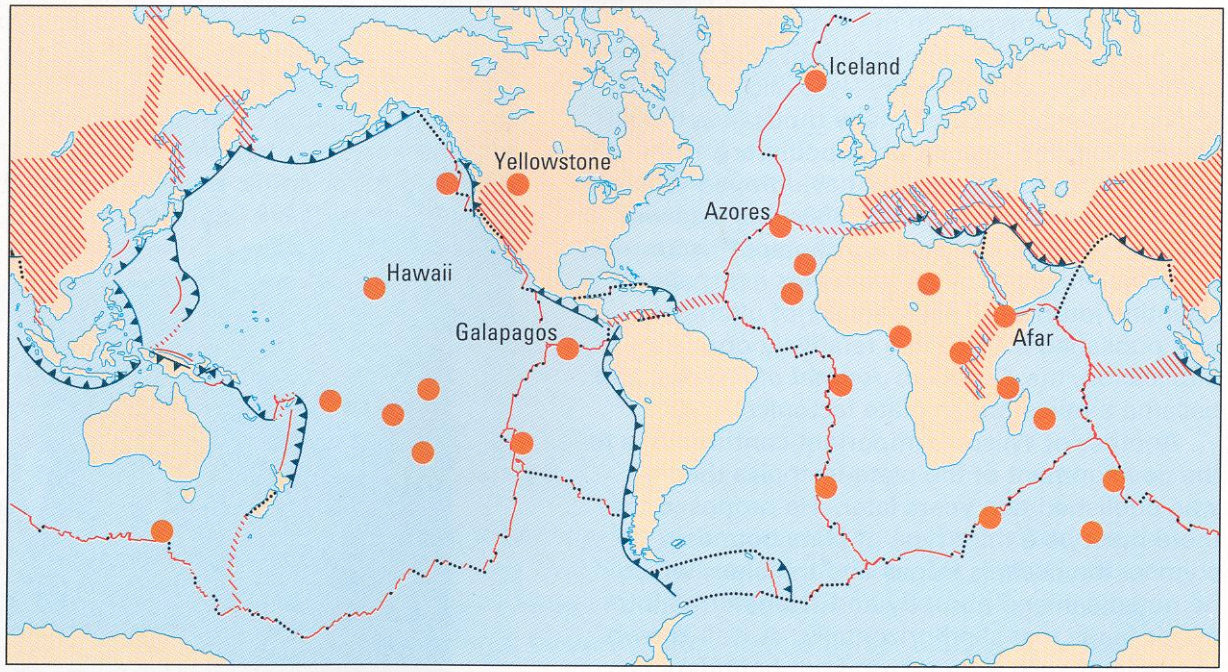
Divergent plate boundaries—Where new crust is generated as the plates pull away from each other.

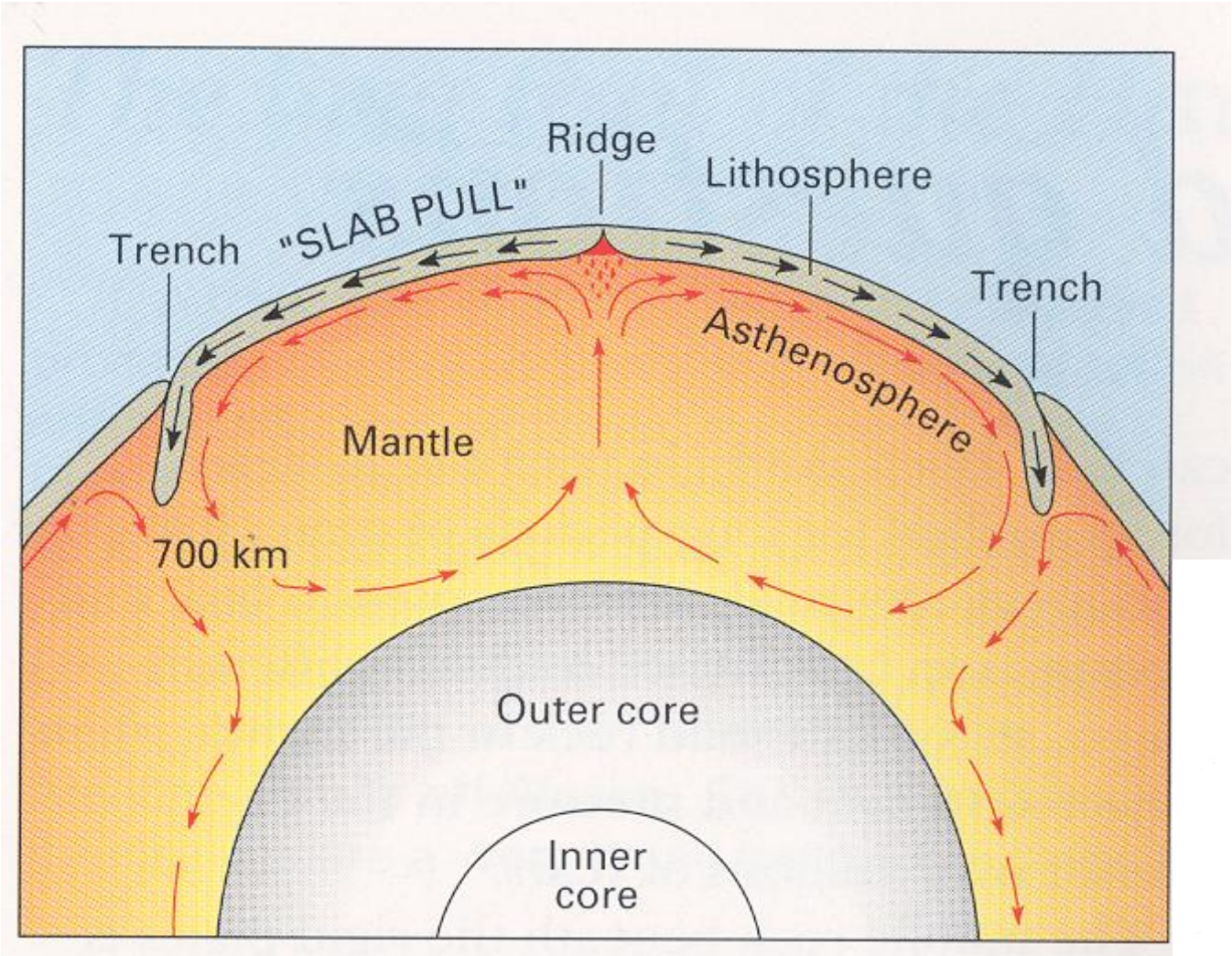
Convergent plate boundaries—Where crust is consumed in the Earth's interior as one plate dives under another.

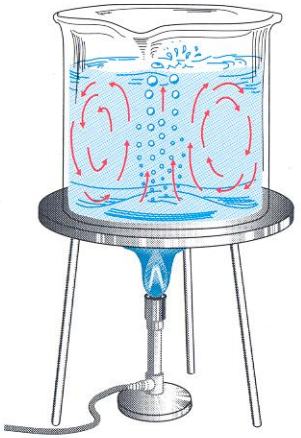
Transform plate boundaries—Where crust is neither produced nor destroyed as plates slide horizontally past each other.

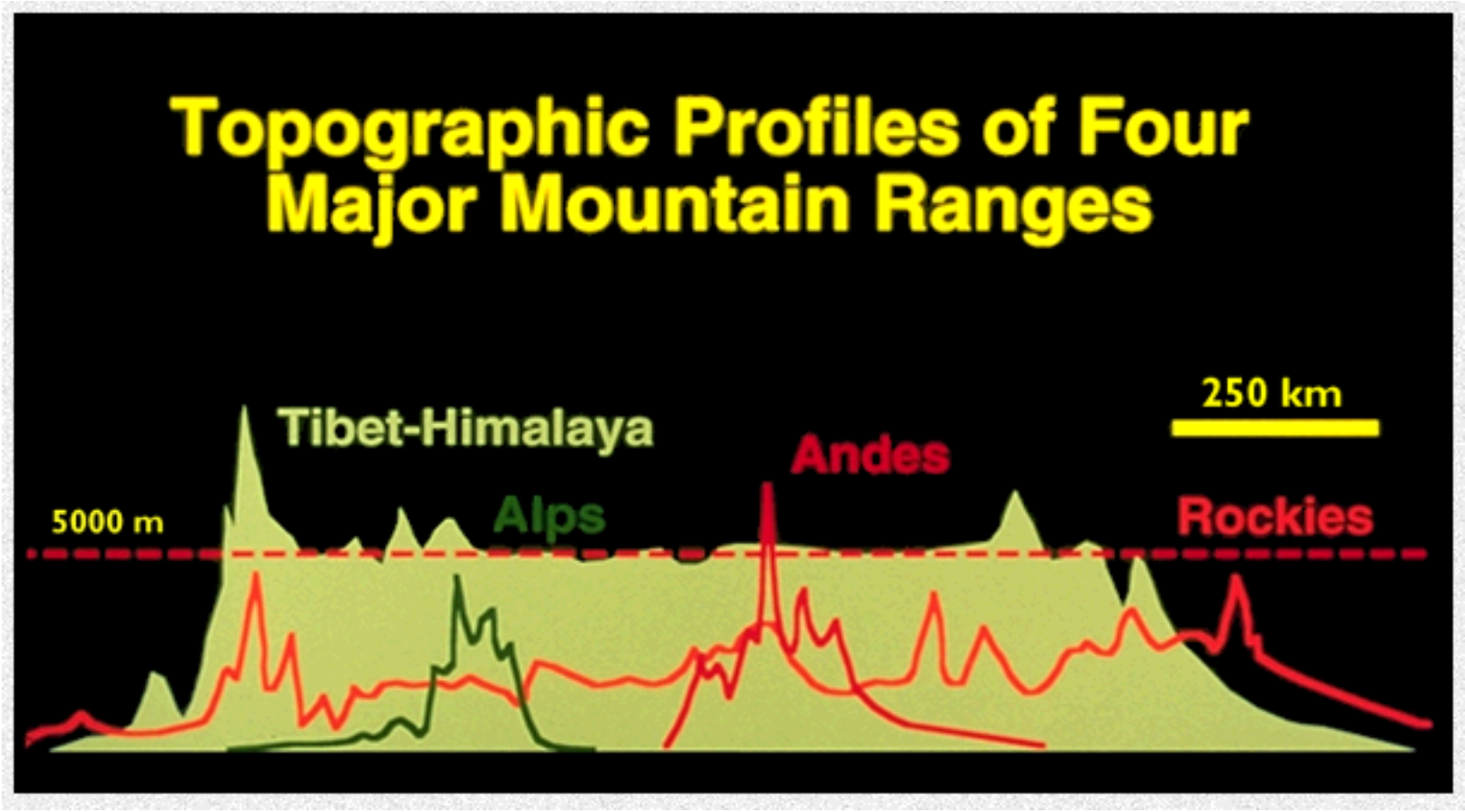
Plate boundary zones—Broad belts in which deformation is diffuse and boundaries are not well defined.

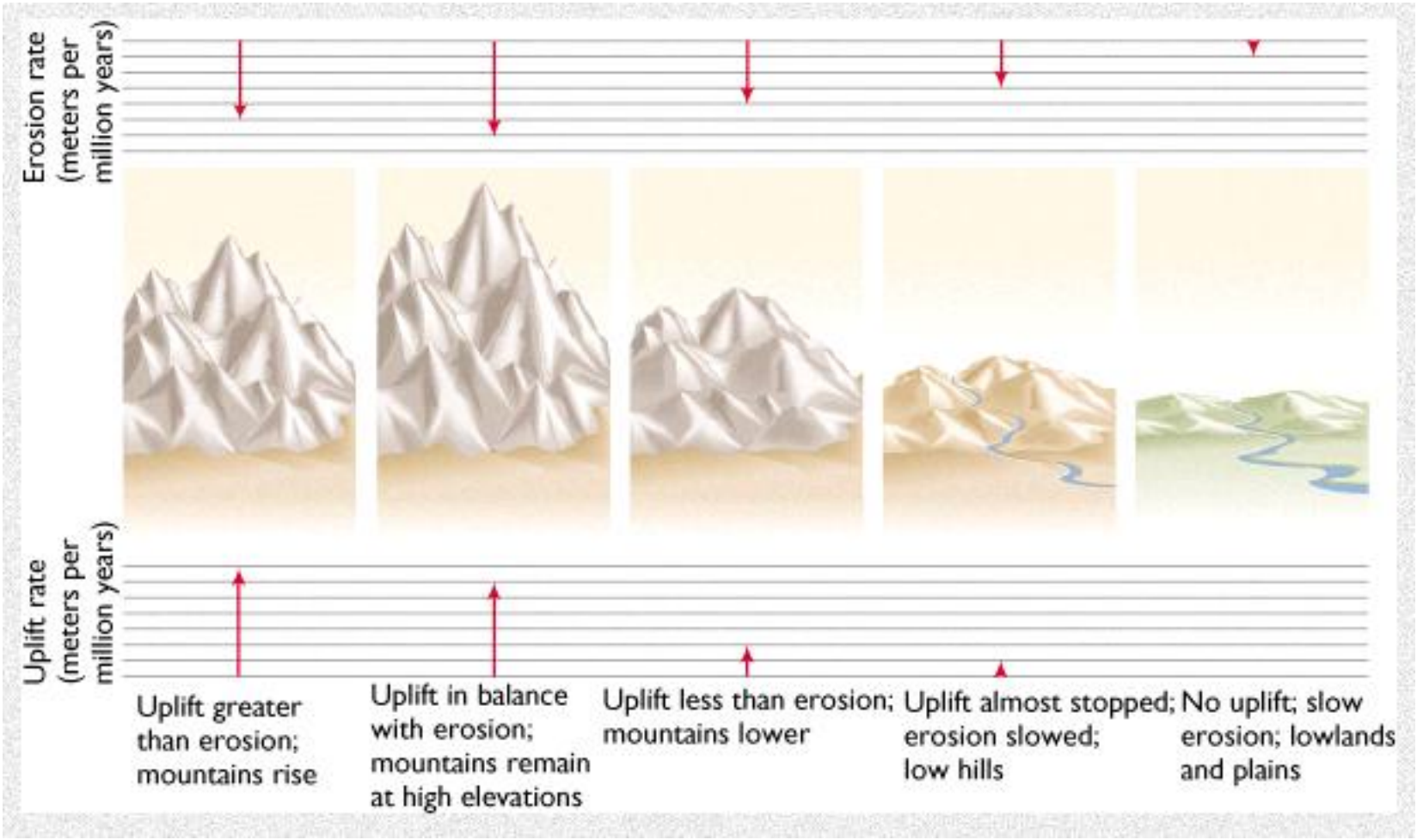
Selected prominent hotspots

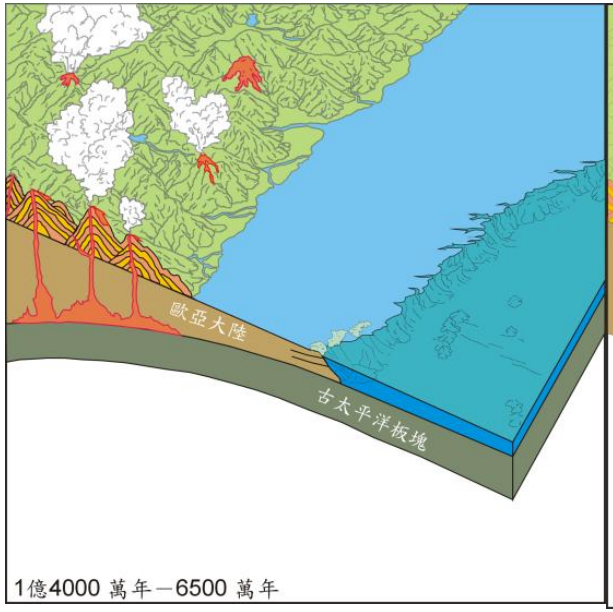



Ridge
Lithosphere
"SLAB PULL"
Trench
Trench
Asthenosphere
Mantle
700 km
Outer core
Inner
core

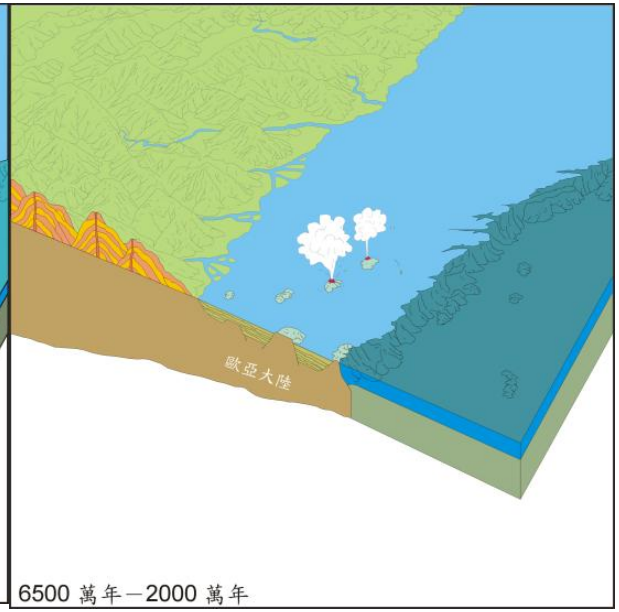



Topographic Profiles of Four Major Mountain Ranges
250 km
Tibet-Himalaya
Andes
Alps
Rockies
5000 m

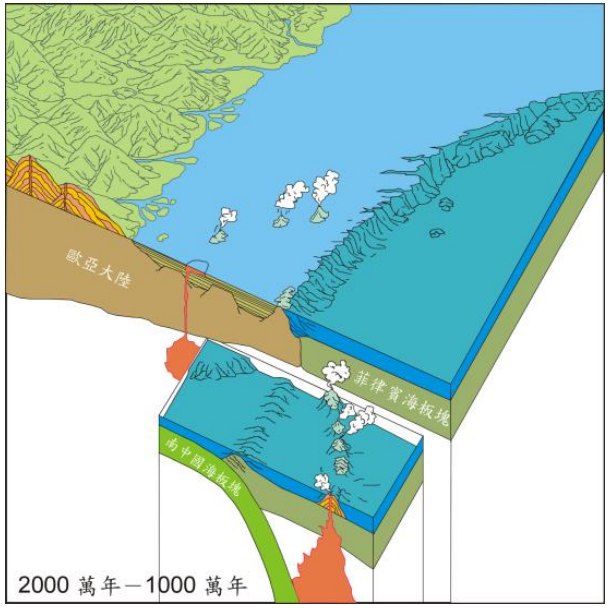

Erosion rate (meters per million years)
Uplift rate (meters per million years)
Uplift greater than erosion; mountains rise
Uplift in balance with erosion; mountains remain at high elevations
Uplift less than erosion; mountains lower
Uplift almost stopped; erosion slowed; low hills
No uplift; slow erosion; lowlands and plains



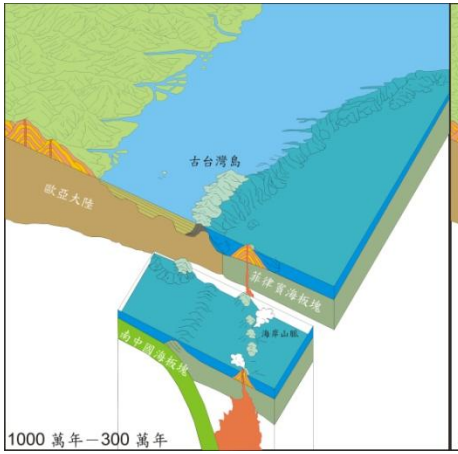
1億4000 萬年－6500 萬年



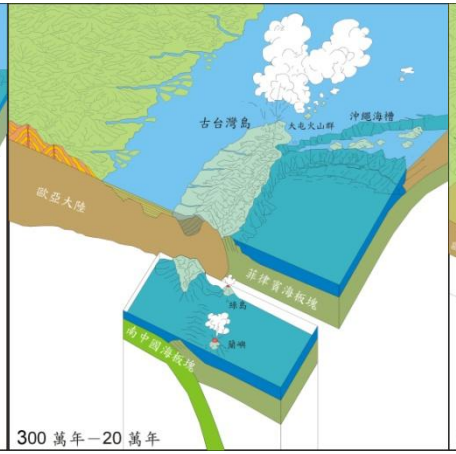
6500 萬年－2000 萬年



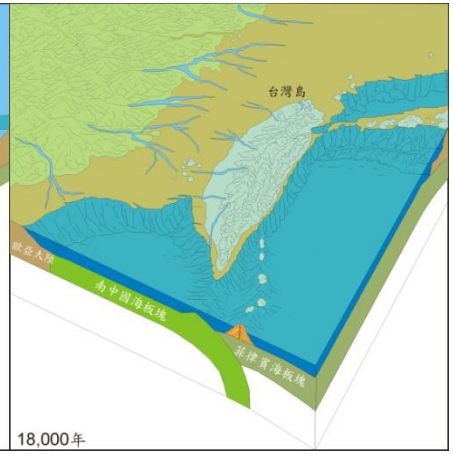
2000 萬年－1000 萬年



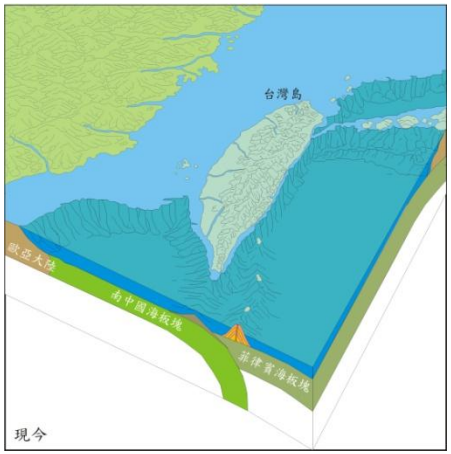
1000 萬年－300 萬年



300 萬年－20 萬年



18,000年



現今



An 18 km-high volcanic plume from one of a series of explosive eruptions of Mount Pinatubo beginning on 12 June 1991, viewed from Clark Air Base (about 20 km east of the volcano). Three days later, the most powerful eruption produced a plume that rose nearly 40 km, penetrating well into the stratosphere. (Photograph by David H. Harlow, USGS.)



Geothermal powerplant at The Geysers near the city of Santa Rosa in northern California. The Geysers area is the largest geothermal development in the world. (Photograph by Julie Donnelly-Nolan, USGS.)